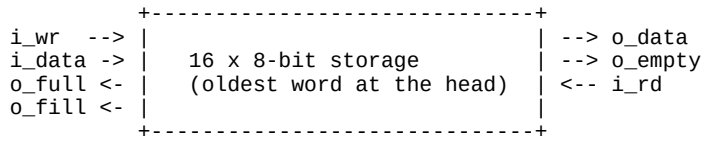
Synchronous FIFO: block specification

1. Overview

`sfifo` is a single-clock first-in, first-out buffer. A producer pushes words with `i_wr`, a consumer pops them with `i_rd`, and the FIFO reports how full it is. The read side is first-word fall-through: whenever the FIFO is not empty, `o_data` already shows the oldest word, and `i_rd` removes it.



2. Parameters

Name	Default	Meaning
BW	8	Data width in bits
LGFLEN	4	log2 of the depth; the FIFO holds 2^LGFLEN words
OPT_ASYNC_READ	1	1 reads storage combinationally; 0 uses a registered read path
OPT_WRITE_ON_FULL	0	1 lets a write proceed when full if a read happens in the same cycle
OPT_READ_ON_EMPTY	0	1 lets a word written into an empty FIFO pass straight through in the same cycle

The verification target is `BW=8, LGFLEN=4, OPT_ASYNC_READ=0, OPT_WRITE_ON_FULL=0, OPT_READ_ON_EMPTY=0`: 8-bit words, depth 16, registered read path, and neither pass-through option. The requirements below describe that configuration.

3. Ports

Port	Dir	Width	Description
i_clk	in	1	Clock. All inputs are sampled on the rising edge.
i_reset	in	1	Synchronous reset, active high
i_wr	in	1	Write request: push `i_data` at this clock edge
i_data	in	BW	Write data
o_full	out	1	High when the FIFO holds 2^LGFLEN words

Port	Dir	Width	Description
o_fill	out	LGFLEN+1	Number of words currently held (0 to 2^LGFLEN)
i_rd	in	1	Read request: pop the word on `o_data` at this clock edge
o_data	out	BW	Oldest word in the FIFO. Valid whenever `o_empty` is low.
o_empty	out	1	High when the FIFO holds no words

4. Functional requirements

Capacity and status

• **REQ-CAP-1** The FIFO holds up to 2^LGFLEN words (16 with the target parameters). It accepts exactly 16 writes from empty before reporting full.

• **REQ-STAT-1** `o_fill` equals the number of words held: the number of accepted writes minus the number of accepted reads since reset.

• **REQ-STAT-2** `o_empty` is high exactly when `o_fill` is 0.

• **REQ-STAT-3** `o_full` is high exactly when `o_fill` is 2^LGFLEN.

• **REQ-STAT-4** `o_fill`, `o_empty` and `o_full` change only on a clock edge: an accepted write or read at one edge is reflected in the status outputs for the following cycle, never earlier in the same cycle.

Writes

• **REQ-WR-1** A write is accepted when `i_wr` is high and `o_full` is low at a rising clock edge. The word on `i_data` becomes the newest entry.

• **REQ-WR-2** A write while the FIFO is full is ignored: no stored word changes, `o_fill` stays at 2^LGFLEN, and no data is lost or overwritten. This holds even when `i_rd` is high in the same cycle.

Reads

• **REQ-RD-1** Whenever `o_empty` is low, `o_data` holds the oldest word in the FIFO (first-word fall-through). No read request is needed to see it.

• **REQ-RD-2** A read is accepted when `i_rd` is high and `o_empty` is low at a rising clock edge. The oldest word is removed and, if words remain, `o_data` shows the next oldest one in the following cycle.

• **REQ-RD-3** A read while the FIFO is empty is ignored: `o_fill` stays at 0 and later data is not affected.

Ordering

• **REQ-ORD-1** Words are read out in exactly the order they were accepted. No word is lost, duplicated or corrupted, including when the internal storage wraps around after many writes.

Simultaneous read and write

• **REQ-SIM-1** When the FIFO is neither empty nor full and both `i_wr` and `i_rd` are high, both are accepted at the same edge: the oldest word is removed, the new word is appended, and `o_fill` is unchanged.

• **REQ-SIM-2** When the FIFO holds exactly one word and both `i_wr` and `i_rd` are high, the held word is removed and the new word becomes the only entry. In the next cycle `o_empty` is low and `o_data` is the new word.

• **REQ-SIM-3** When the FIFO is empty and both `i_wr` and `i_rd` are high, only the write is accepted (see REQ-RD-3).

• **REQ-SIM-4** When the FIFO is full and both `i_wr` and `i_rd` are high, only the read is accepted (see REQ-WR-2); `o_full` stays high during that cycle.

Reset

• **REQ-RST-1** While `i_reset` is high at a clock edge, the FIFO is emptied: in the next cycle `o_fill` is 0, `o_empty` is high and `o_full` is low. Writes and reads in a reset cycle are ignored.

• **REQ-RST-2** Reset may be applied at any time, including when the FIFO holds data. After reset the FIFO behaves exactly like a freshly reset FIFO: words written afterwards are read back in order and no word from before the reset ever appears on `o_data`.

5. Out of scope

Separate read and write clocks, almost-full/almost-empty thresholds and the non-default option settings are not part of this verification target. The contents of `o_data` while `o_empty` is high are undefined.